

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANASANGAMA, BELAGAVI – 590018



An Internship Report
On
JAVA APPLICATION USING SPRING AND HIBERNATE

Submitted in partial fulfillment for the award of degree of

Bachelor of Engineering
In
Computer Science and Engineering

Submitted by
R Narendranath
Reddy
1BG17CS075



Vidyayāmruathamashnuthe

B.N.M. Institute of Technology

Approved by AICTE, Affiliated to VTU, Accredited as grade A Institution by NAAC.
All UG branches – CSE, ECE, EEE, ISE & Mech.E accredited by NBA for academic years 2018-19 to 2020-21 & valid upto 30.06.2021
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Department of Computer Science and Engineering

2020-21

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Department of Computer Science and Engineering



Vidyayāmṛuthamashnute

CERTIFICATE

Mr. R Narendranath Reddy USN **1BG17CS075**, a bonafide student of VIII Semester, Department of Computer Science and Engineering, **B.N.M Institute of Technology** has been successfully completed Internship at **Glisten Project Solutions Private Limited** on **Java Application using Spring and Hibernate** in partial fulfillment for the award of Bachelor of Engineering in COMPUTER SCIENCE AND ENGINEERING of **Visvesvaraya Technological University**, Belagavi in the year 2020-2021. The Internship report has been approved as it satisfies the academic requirements with respect to the Internship work prescribed for the said degree.

Prof. Priyanka Padki
Assistant Professor
Department of CSE
BNMIT, Bengaluru

Dr. Sahana D. Gowda
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Department of CSE
BNMIT, Bengaluru

Dr. Krishnamurthy G
Principal
BNMIT, Bengaluru

Name & Signature

Examiner 1:

Examiner 2:



Certificate of Internship

THIS CERTIFICATE IS PRESENTED TO

Mr./Ms./Mrs. - R Narendranath Reddy

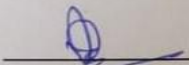
Intern ID: GPSPL - 163 USN ID: 1BG17CS075

in appreciation of work as intern

Course: Java with spring and hibernate

Project: Online cab Booking system

From Date: 7th June 2020 To Date: 7th August 2020


Mentor
Glisten Projects

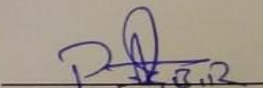

Internship Head
Glisten Projects

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The success and final outcome of this internship report required a lot of guidance and assistance from many people and I am extremely privileged to have got this all along the completion of my internship.

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R Narendranath Reddy

1BG17CS075

Chapter 1

ABOUT THE COMPANY

1.1 Introduction

Glisten Project Solutions Private Limited is one of the leading training institutes in Bangalore, Mysuru, Kolar, Tumkuru and it is among top IT Educational companies. Established in the year 2011, since its inception, it has been growing at an exponential rate. It has 7 centre's in Bangalore, one each in Mysuru, Kolar, and Tumkuru.

Glisten Project Solutions Private Limited is a Private incorporated on 03 December 2013. It is classified as Non-govt company and is registered at Registrar of Companies, Bangalore. Its authorized share capital is Rs. 100,000 and its paid up capital is Rs. 1,00,000. It is involved in Business activities.

Glisten Project Solutions trains Corporate Clients, Academies and individuals in the domains of Software, Web development, Embedded Systems, IT infrastructure, Digital marketing, Tally, CADD (Civil, Mechanical & Electrical), Aptitude (General & Technical) & Soft skills. Glisten Education is the training and placement vertical which is one among the finest Institutions in Bengaluru the provides software training with placement assistance, Cost-Effective, Quality and real time training courses and projects of Autodesk, Adobe, GCADD, Hewlett Packard Enterprise, Microsoft and IIHT.

It also trains on the latest of technology advancements in the industry, like Finance, Accounts, SAP, Cloud Computing, Networking, AI, IoT, Big-data, designing concepts, Block Chain, 3D printing technology, etc.



Figure 1.1 Glisten Logo

1.2 Services

Services offered are in the fields of Engineering, Management, Networking and Transformations.

- **Software Training:**

Software training specialists focus on teaching people how to use software, so they need strong computer skills. They also need motivational skills so that they can engage students, as well as communication skills so that their training materials are effective and easy to understand.

- **Networking and Hardware**

Hardware and networking components are critical to the functioning of the IT industry and other verticals. Complementing each other functionally, data and information is distributed by linking group of networks using hardware components like switches, interface cards and networking cables.

- **CADD Training and Projects**

AutoCAD training programs consist of only one comprehensive course that includes progressive instruction from basic to advanced skill levels.

1.3 Vision

- Be a globally recognized training Institute in the field of Engineering and Information Technology Education.
- Facilitate educational access and academic excellence through exceptional management of operations and resources, innovative delivery systems, student services and relevant academic programs that are success oriented and responsive to technology.

Chapter 2

ABOUT THE DEPARTMENT

2.1 Introduction

Web has a dictionary meaning of a complex system of interconnected elements. The true character of its name is it's the Google created software stack for creating comprehensive desktop and mobile web applications and software to realize the full potential of application's end user. Web programming, also known as web development, is the creation of dynamic web applications. Examples of web applications are social networking sites like Facebook or e-commerce sites like Amazon.

Web is a comprehensive software stack of desktop and mobile application that includes operating systems, middleware and key application. Web initially came into existence with the fire idea that developments are given the freedom to create enthralling Web applications while taking advantage of everything that the operating system of PCs and mobiles has to offer. Applications are developed for variety of reasons: addressing and business requirements, building new services, creating new business and other types of content for users. Developer choice to develop for web in order to reach the majority of end users. To develop the applications efficiently, there are many ways of addressing based on the development environment. The web page can be built either only front-end or both frontend and back-end based on the requirement.

2.2 Web Technologies

The front-end development takes place with the help of Hyper Text Markup Language or simply HTML in the core and the styling can be done using Cascading Style Sheet or CSS. The requirements of the application decide whether it requires a back-end or not. The backend is generally a database, which may be structured or unstructured depending on the type of data to be stored. And the PHP or Hypertext Preprocessor is a widely-used open source general-purpose scripting language that is especially suited for web development and can be embedded into HTML.

The PHP acts as the middleware connector to bind the front-end and the back-end. There are many other web development scripting languages that can be embedded in HTML.

Some of them are: Angular JS, Node JS, React JS, Java Script, etc., which makes the web application more responsive.

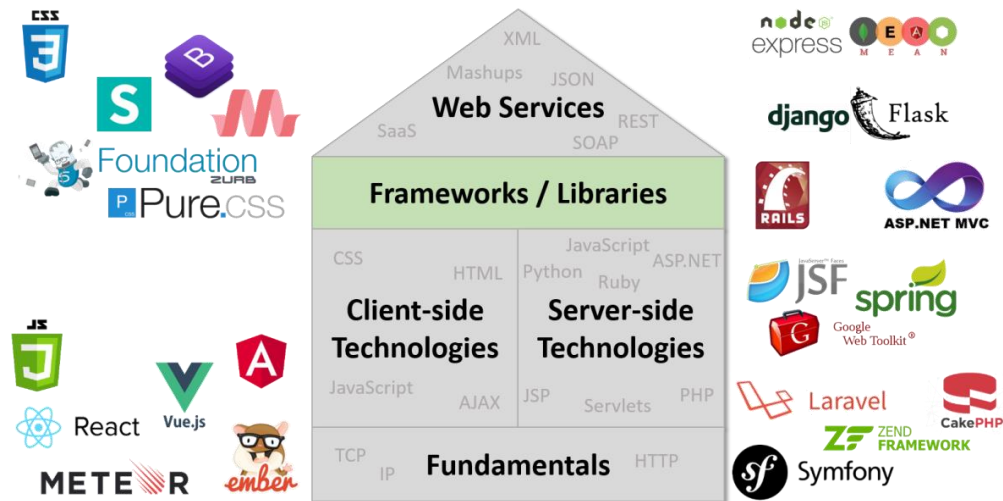


Figure 2.1 Web Technologies

2.3 Spring and Hibernate in Java

Spring and Hibernate are frameworks that are used in Java. Spring is used to develop application from desktop to Web. Hibernate is used to access data layer and Struts is used for Web frameworks.

2.3.1 Spring

Spring framework is used to develop software application. It has lot of components at the same time it provides different functionalities. Spring Components are :

- Spring Model View Controller (MVC): MVC helps for app development.
- Spring Core: It provides Inversion of Control (IOC) or dependency Injection.
- Spring Transaction Management: It provides transaction management for apps.
- Data Access Framework: It provides data or information management functionality.

2.3.2 Hibernate

Hibernate is an Object-Relational Mapping (ORM) framework. It uses Java Data Base Connectivity (JDBC). It provides flexibility to change the database. Java Persistence API (JPA) are also used, ORM like Toplink, Eclipselink, etc.

Chapter 3

TASKS PERFORMED

3.1 Web Development

During internship, the intern was exposed to various domains of the work culture and was given a chance to showcase the interest towards the dedicated project or application. Some of the software's and development languages that has been used in this project are,

- **HTML5**

HTML5 is a software solution stack that defines the properties and behaviors of web page content by implementing a markup based pattern to it. HTML5 is the fifth and current major version of HTML, and subsumes XHTML. The current standard, the HTML Living Standard is developed by WHATWG, which is made up of the major browser vendors (Apple, Google, Mozilla, and Microsoft), with the Living Standard also existing in an abridged version.

- **CSS**

CSS Stands for "Cascading Style Sheet." Cascading style sheets are used to format the layout of Web pages. They can be used to define text styles, table sizes, and other aspects of Web pages that previously could only be defined in a page's HTML. CSS helps Web developers create a uniform look across several pages of a Web site. Instead of defining the style of each table and each block of text within a page's HTML, commonly used styles need to be defined only once in a CSS document.

- **MySQL**

MySQL is a relational database management system based on SQL – Structured Query Language. The application is used for a wide range of purposes, including data warehousing, e-commerce and logging applications. The most common use of MySQL however, is for the purpose of a web database. It can be used to store anything from a single record of information to an entire inventory of available products for an online store.

- **Java Sever Page**

Java Server Pages (JSP) is a server-side programming technology that enables the creation of dynamic, platform-independent method for building Web-based applications. JSP have access to the entire family of Java APIs, including the JDBC API to access enterprise databases.

- **Hibernate Query Language**

Hibernate Query Language (HQL) is an object-oriented query language, similar to SQL, but instead of operating on tables and columns, HQL works with persistent objects and their properties. HQL queries are translated by Hibernate into conventional SQL queries, which in turns perform action on database.

- **Eclipse IDE**

Eclipse is an integrated development environment (IDE) for developing applications using the Java programming language and other programming languages such as C/C++, Python, PERL, Ruby etc.

- **Tomcat Server**

Apache Tomcat, also known as Tomcat Server, proves to be popular choice for web developers building and maintaining dynamic websites and applications based on the Java software platform.

3.2 Implementation

The application is primarily designed to book cabs between an entered source and destination along with vehicle of our choice by logging on to the website. It has all the basic facilities that are required by a user to have a convenient and hassle-free online booking experience.

Model-View-Controller (MVC) architecture provides the simplest way to develop flexible and loosely coupled web applications. It is a design pattern that separates the business logic, presentation logic and data.

- Model Represents the state of the application

- View represents the presentation
- Controller acts as an interface between view and model.

At the front, the user login and the registration page is created by configuring the dependency settings in XML page with the implementation of controller class. The dynamic and interactive webpages are created using JSP. The user's homepage facilitates the user to input the start and the destination journey. It subsequently displays the list of all available cab services for the intended journey from the MySQL database using the Hibernate ORM. The request to the desired cab service provider is sent uninterrupted. Once the request is accepted, a confirmatory message is notified in the user's portal along with a detailed mail of confirmation sent to the associated user. Concurrently the status is updated in the database. The feature of 4-digit OTP based verification is also implemented that eliminates all the flaws in the process and thereby ensuring a safer journey. This application development includes:

- spring-web and the spring- webmvc module that we need for our Spring web application
- a Tomcat starter so that we can run our web application directly without explicitly installing any server.

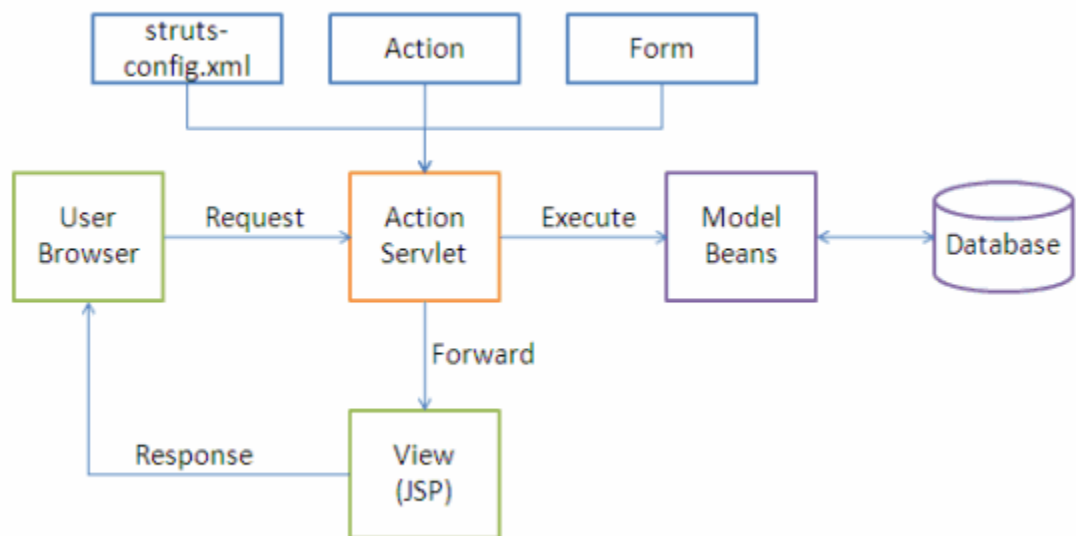


Figure 3.1 MVC Architecture

3.3 Modules Involved

1) ADMIN

The below given are four functionalities performed by the Admin.

- **Manage Dispatcher**

An admin can manage the dispatcher who can handle booking requests. It can assign the roles to dispatchers and operators. The dispatcher can manage the drivers' tasks accordingly.

- Assign Dispatcher roles
- Add/modify delete dispatchers
- View and track dispatcher activities

- **Manage Passengers**

Admin is responsible for managing the passengers registered on the system. Admin can add/view/modify/remove the passenger details anytime.

- View passenger details
- View booked trips details
- Edit details
- Add/delete passenger details

- **Manage Drivers**

Admin is responsible for managing the drivers registered on the system. Admin is able to add/modify /delete drivers anytime. It can accept or reject a driver's account. Only admin approved driver accounts can get online.

- Review vehicles and documents added by drivers
- Approve/reject driver's account
- Add vehicle type

- Add driver vehicles

- **Reporting Analytics**

A report section where Admin can view various reports like Total Earning Report, Payment Report, User wallet Report, Referral Report, Drive Log and Payment Report etc.

- Check key insights and numbers
- Check how much you are earning
- Check whether to increase/decrease commission

2) DISPATCHER

A thoroughly designed dispatch system to easily manage all the dispatch requests with real-time tracking and different dispatch modes.

- **Driver Management**

A dispatcher can add drivers, review signup requests, set commission rates and more. They can approve or reject a driver permit by verifying if it's valid. Also, they can suspend a driver anytime.

- Assign trips to drivers
- Manage taxi locations
- View driver details
- Alarm and alerts for driver safety

- **Booking Management**

The taxi dispatch system works according to your existing web taxi booking module. It can control the incoming bookings and dispatch them. It will have an overview of how drivers and rides are functioning on a daily basis.

- Dispatcher Integration
- Customize booking
- Assign Drivers to requests

- **Trips Tracking**

A dispatcher can track every vehicle in real-time, whether it's idle or in dispatch, with the integrated GPS tracking system. They can track the current location of the drivers including the fleet availability and assign the jobs accordingly.

- Dedicated GPS
- Live location tracking
- View and track real-time fleet availability
- View and monitor trip history

- **Manual Dispatching**

Dispatchers can create new bookings for phone calls and use auto assignment. In this way, the dispatcher panel can receive requests manually. While manual dispatching is required in exceptional cases, auto dispatching is used in most of the cases.

- Easy switching between the modes
- In-app support
- Receive request through phone calls
- Manually assign a driver to the provider

- **Emergency Management**

For the safety of passengers and drivers, the dispatcher will be able to view all the emergency requests for a safe and secure journey and ensure quick actions if an emergency occurs during the ride.

- Call drivers
- Quick actions during an emergency
- Share details with concerned authorities

3) CUSTOMER

The Online Cab Booking System includes features which provides a flawless and seamless experience to our customers.

- **Pick the Destination**

Enter the first few words and the app will suggest the destination.

The app auto detects your passenger's current location. All he has to enter is where he wants to go to book a taxi and reach his destination. No need to give directions to the driver.

- Assisted Search for your destination

- Get the estimated fare
- Pin locations you frequently commute to
- Get your route detail on the in-app map

- **Pin location**

Save locations you frequently travel to.

Pin destination you travel to regularly and name them as Home, Office, Friend's place, etc and un-complicate the entire process of booking a taxi. As soon as you proceed to book, pinned locations will populate for quick booking.

- Pin locations you frequently commute to
- Get your route detail on the in-app map
- Rename pinned locations' name
- The quickest way to book a taxi.

- **Select Car Type**

Sedan, SUV or a hatch, every customer has different preferences.

You can configure the app to let your customers choose between the types of vehicles they want as their ride. It could be a long, luxurious S Class or a small, compact Mini Cooper.

- Select a Mini, hatch, sedan or ...
- Select car-type in an easy to use app slider
- The driver's name and photo displays
- The driver receives notification

- **Know your driver**

Is he the same person you booked taxi with?

Passenger can learn what is the brand and model of incoming car, how driver looks like, what the number plate says and how far is he from the pickup point. In addition, you can track his real-time location on an interactive map.

- Display driver's name and photo post booking
- Passenger can call or text them from within the app
- Passenger can check a driver's average rating
- Watch him on a map as they drive in your direction

- **Easy Payment**

They can pay with cash or make the trip cashless and pay within app.

Passengers have many payment options to choose from: Cash, Credit Card, depending upon payment gateway you integrate. Time and distance determines the price.

- **Rate Driver**

A driver's average rating should give you a fair idea how they are doing their job.

Riders appreciate when drivers show great customer service. Customer can rate a ride on a scale of 1- 5. Where 1 means terrible, 5 means awesome and 3 means average.

- Rate driver on a scale of 1 to 5.
- Customer can also write a review of the ride.
- Driver can also rate the customer.
- Learn which driver is ruining your brand.

3) DRIVER

We developed the app as a true companion for drivers. That is everything a driver cares about from navigating to the pickup location to keep a track of his or her earnings must be accessible from within the app. At the same time, we wanted an approachable, minimalistic app.

- **Trip History and Fare**

The driver can check the trips he took and fare he collected within the app.

Check which trip you last took or those you took over the day or a month along with how much fare you collected, both in individual trips or all over the time or in a particular month or week within the app.

- Driver can check trip history
 - Fare from each trip
 - Total fare collected till today
 - Sort according to weeks or months
- **One-tap trip start**

With a single tap on the app screen, a driver can start or end a trip

When the passenger is on-board, start the trip with the tap of a button and the app will start calculating fare. No complicated procedures, he just have to tap the button on his app screen and proceed to drop the passenger.

- Single tap to start a trip
- Fare calculation upon trip start
- Destination displays on the app
- Keep things simpler for everyone

- **Complete trip in seconds**

The fare is auto-collected as soon as a driver ends the trip from the app

Once a driver has dropped off the passenger to his destination, the driver can complete the trip with a single tap and the fare amount gets displayed on the both the apps. The customers has the option to make the payment in cash or digitally.

- To complete a trip, a driver must tap 'end trip' button
- The fare to be collected is shown at the end of the trip
- Make payment in cash or with your card
- Passenger app too displays the fare post a trip

- **Fare, time and distance**

The driver app displays essential information at the end of each trip.

Upon completing the trip, the driver app displays duration of the trip, distance travelled, and the fare the driver must receive in the form of trip summary. He can tap the individual details in the summary to learn more about the trip he took.

- The app calculates fare, not driver
- Driver can receive payment in cash
- Driver can report a discrepancy
- The fare collected is displayed too

● **Trip Details**

The app displays every piece of information the driver cares about.

Get summary of the trip you last took including how much time you took, taxes, distance, and a trip map. Likewise, the app gives driver the summary of every trip he ever took for his reference or in case a dispute arises.

- Summary of the last trip
- On screen access to ride history
- Fewer disputes and conflicts
- Fine details of trips made

● **Digital payments**

Drivers can stop worrying about keeping change.

With app transactions happening digitally, you driver would spend more time making money than looking for change after each trip. That means he can more trips in a day, and he don't have to struggle when out of change.

- Digital payment options
- Less time between trips
- Transparency
- Accept all major cards

- **Navigation System**

Navigation helps the drivers to reach at the destination with high accuracy and short time.

- **Accept/Reject request**

This feature gives the drivers an authority to accept or reject the customer's request.

3.3 BENEFITS OF ONLINE CAB BOOKING SYSTEM

- **Security**

It's highly secure as it requires only a logged in person to book the cab which prevents misuse of their details by intruders.

- **High customization**

Our solution offers high customization with the provision to add new features and integration which would help you to achieve higher scalability.

- **Notifications integration**

Our Cab Booking software has notification integration which sends email, SMS updates, and notifications on phone for various events.

- **Quick matching algorithm**

The Cab Booking System has a quick matching algorithm which matches the passengers' request with drivers by ensuring utmost optimization.

- **One-touch operation**

With our cab booking system you can book your cab with just a single touch on your phone. Similarly, the driver can accept the request in one touch.

- **Ease of use**

The Cab Booking App is User-Friendly and has precise instructions for both drivers and customers which are easy to understand.

- **Merit-based incentives**

Our Solution provides accurate data which would help you to provide Merit-based incentives to the high performing drivers.

3.4 Results

Following are the screenshots and descriptions of the pages designed.

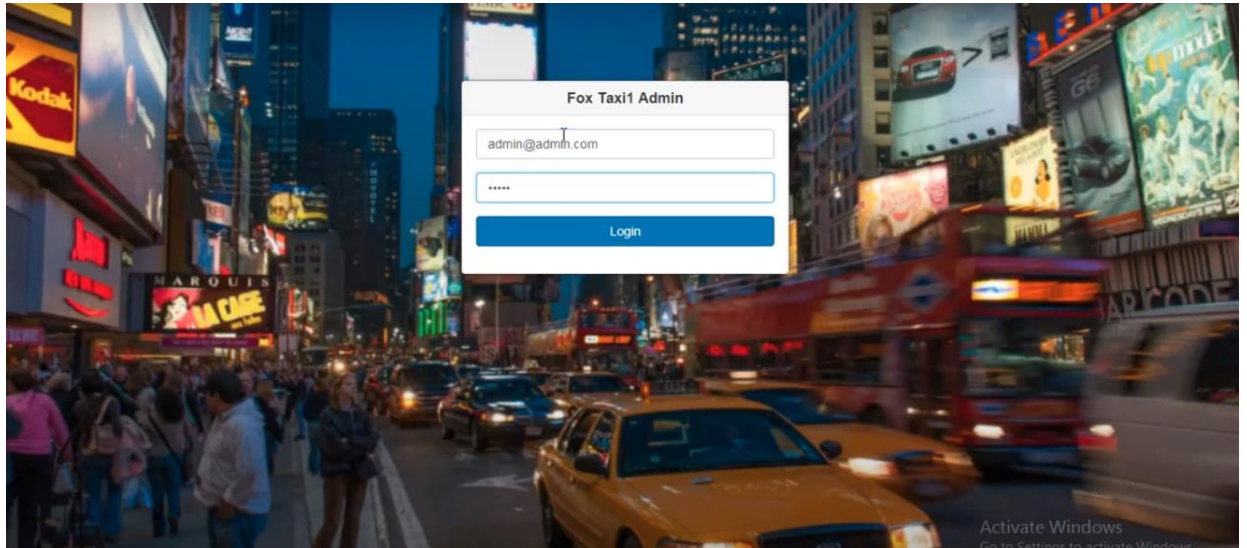


Figure 3.2 Admin Login

Login page of Admin where username and password is authenticated.

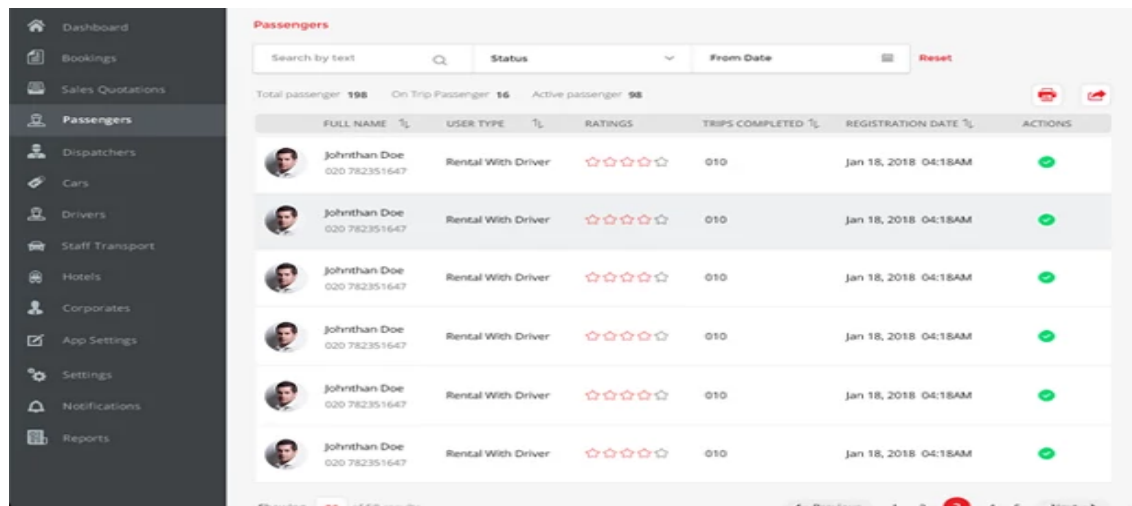


Figure 3.3 Manage Passengers

Admin can add/view/modify/remove the passenger details anytime.

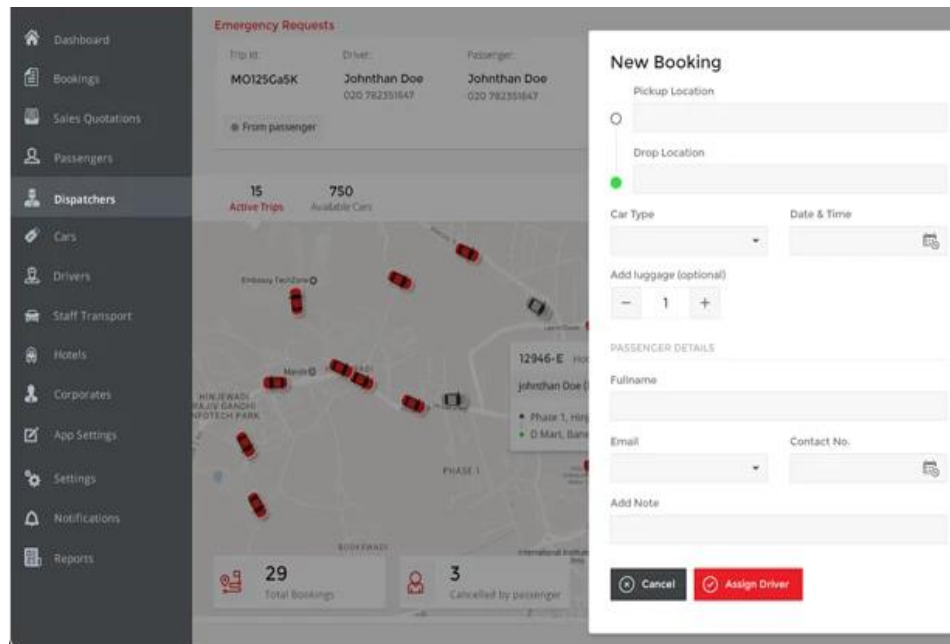


Figure 3.4 Manage Dispatcher

An admin can manage the dispatcher who can handle booking requests.

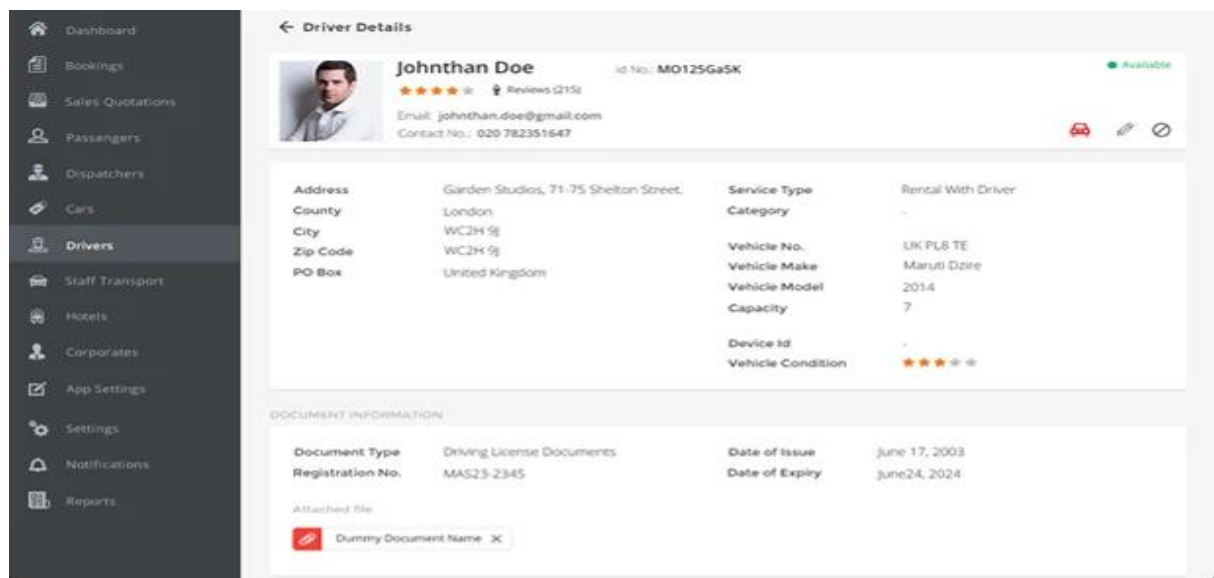


Figure 3.5 Manage Drivers

Java Application using Spring and Hibernate

A report section where Admin can view various reports like Total Earning Report, Payment Report, User wallet Report, Referral Report, Drive Log and Payment Report etc.



Figure 3.6 Reporting Analytics

Admin is able to add/modify /delete drivers anytime. It can accept or reject a driver's account.

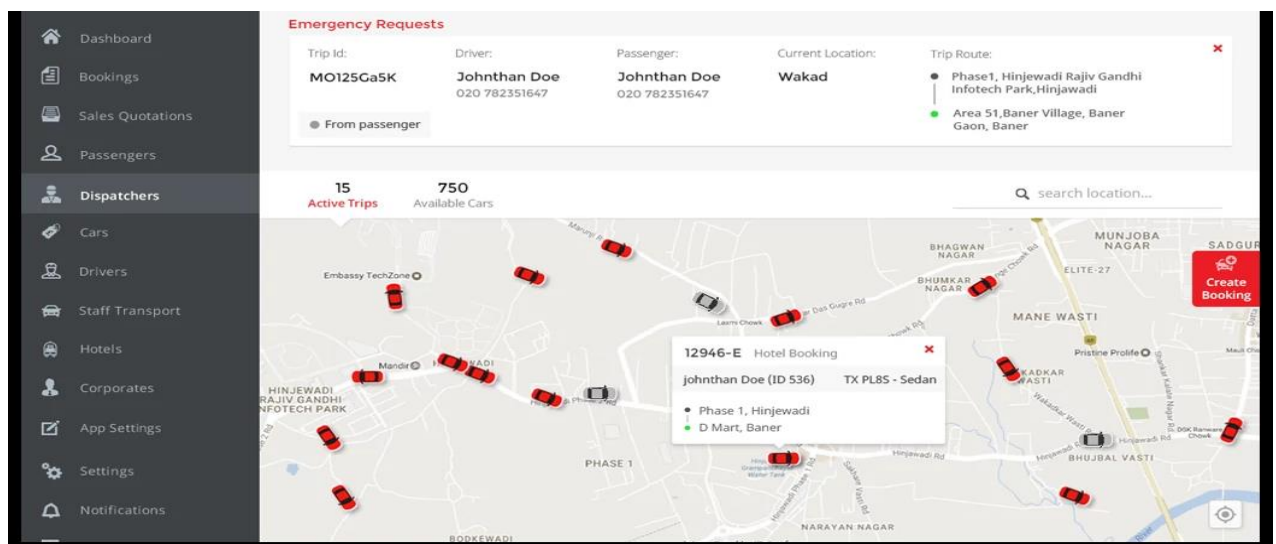


Figure 3.7 Dispatcher Panel

Java Application using Spring and Hibernate

Manages all the dispatch requests with real-time tracking and different dispatch modes.

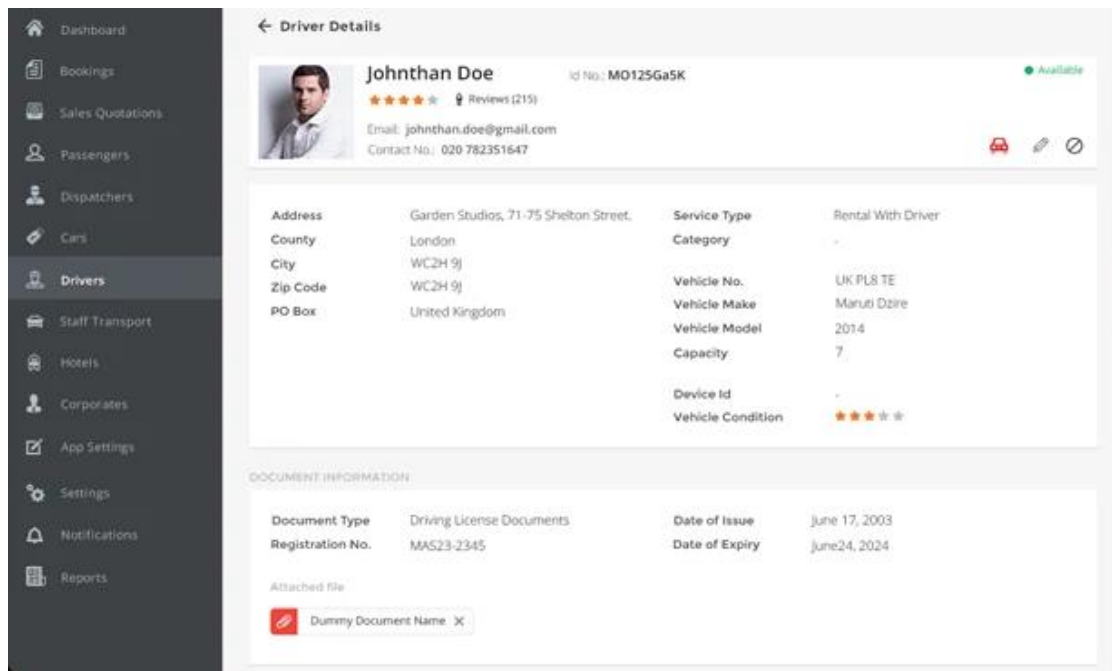


Figure 3.8 Driver Management

A dispatcher can add drivers, review signup requests, set commission rates and can approve or reject a driver permit.

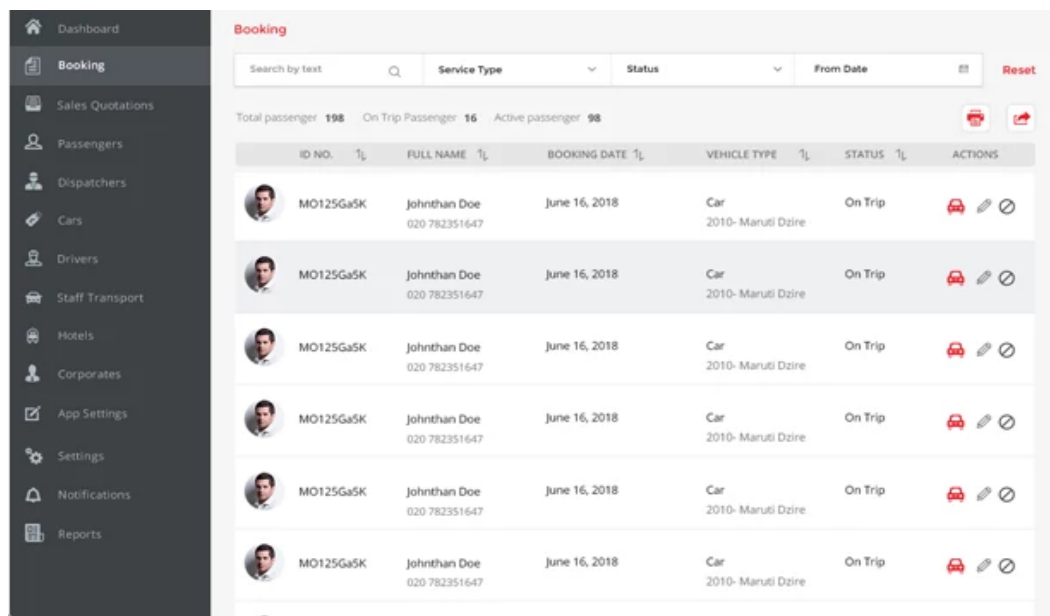


Figure 3.9 Booking Management

It will have an overview of how drivers and rides are functioning on a daily basis.

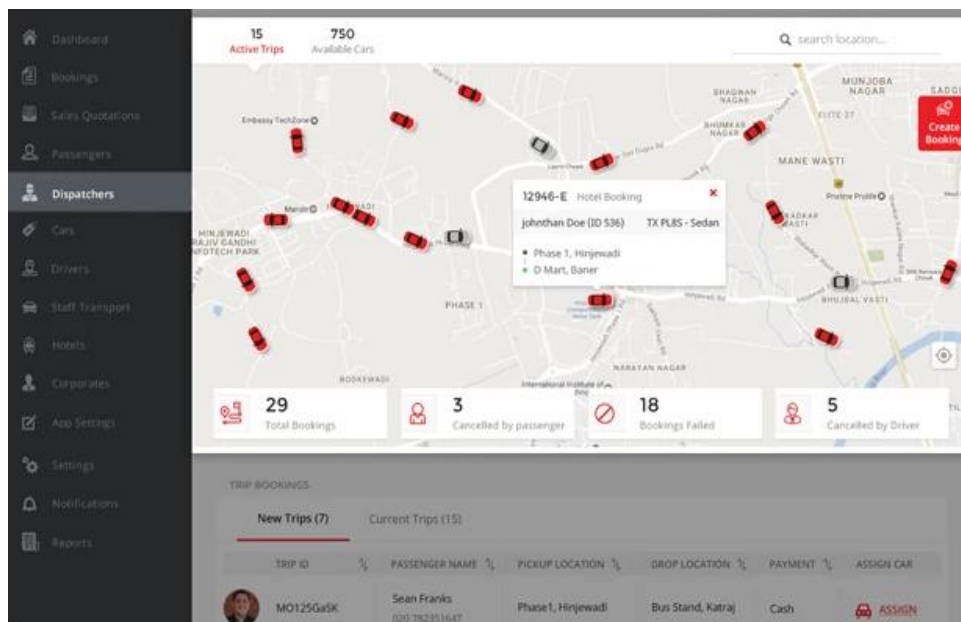


Figure 3.10 Trips Tracking

A dispatcher can track every vehicle in real-time with the integrated GPS tracking system.

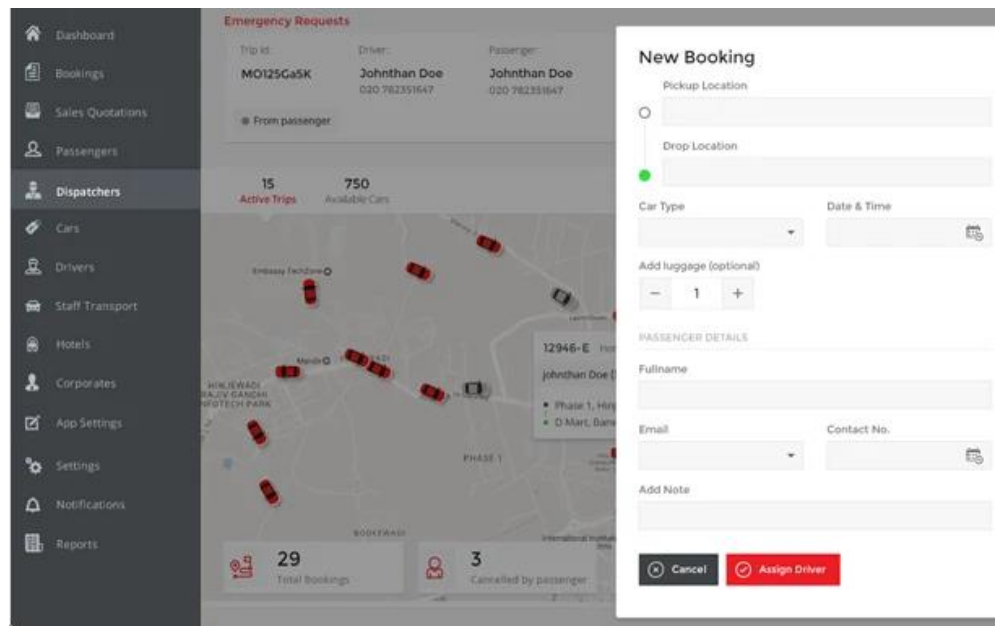


Figure 3.11 Manual Dispatching

Manually assign a driver to the provider through phone calls.

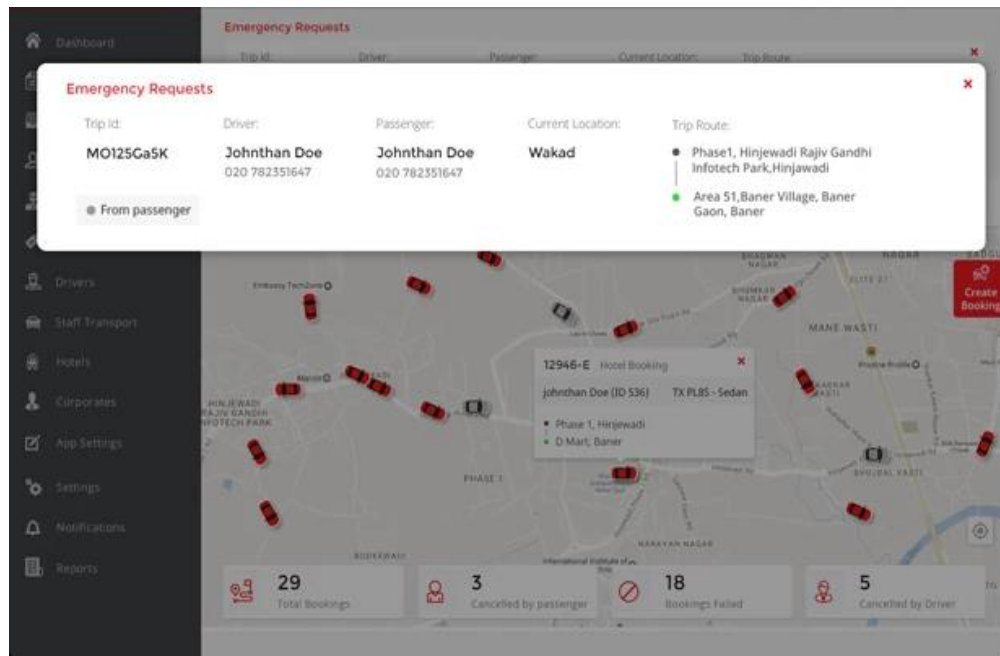


Figure 3.12 Emergency Management

The dispatcher will be able to view all the emergency requests for a safe and secure journey and ensure quick actions if an emergency occurs during the ride.

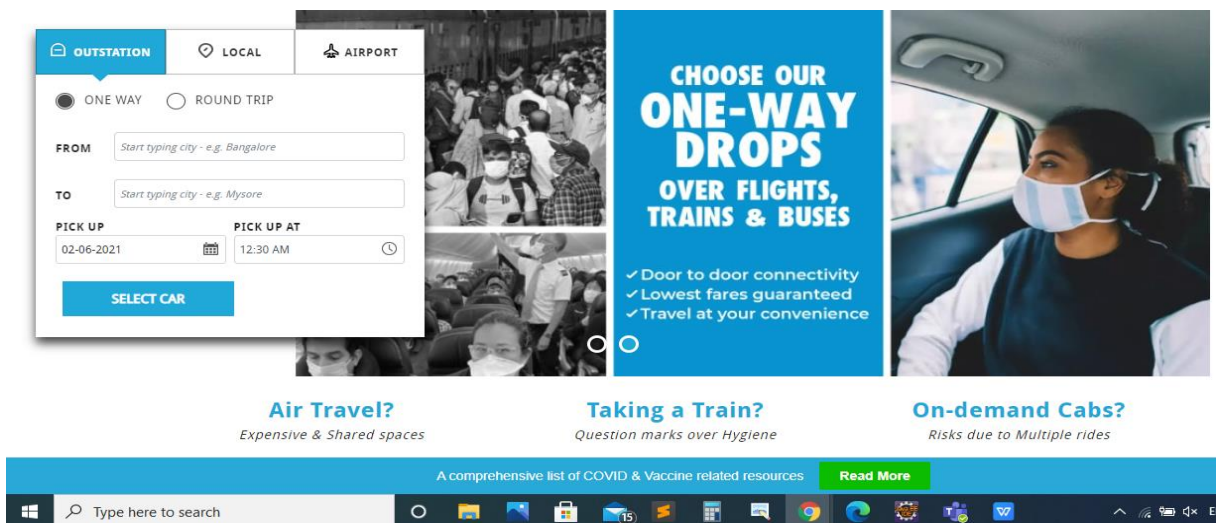


Figure 3.13 Customer Panel

Get a customized UI to personalize the cab booking experience.

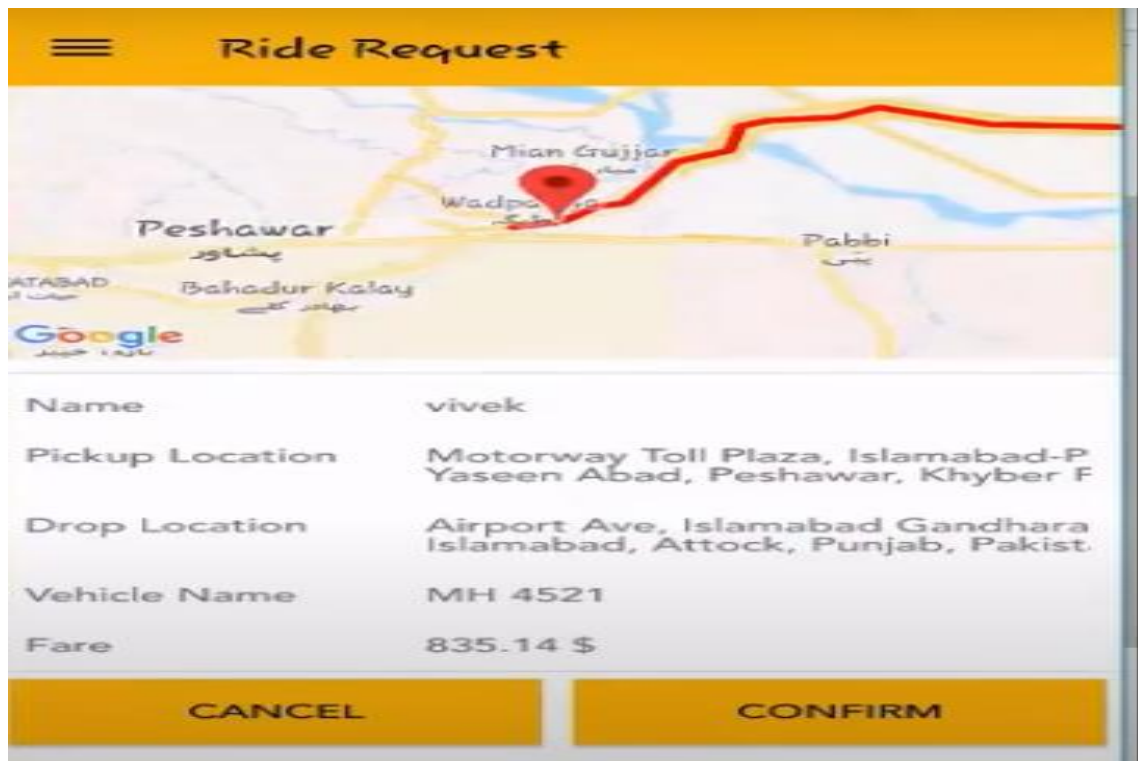


Figure 3.14 Customer Bookings

Customers can register and login through multiple methods such as social media login, or create an account.

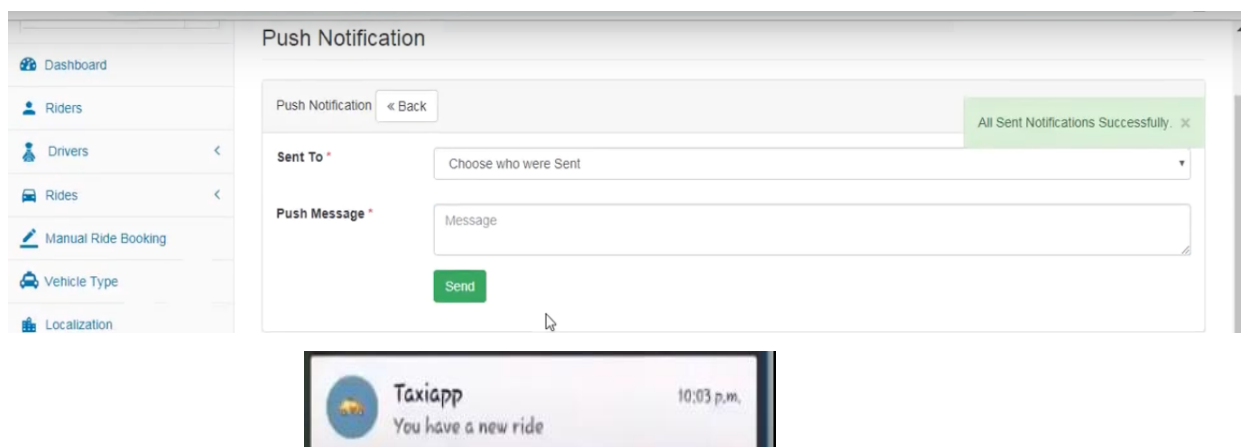


Figure 3.15 Notifications

The passengers can get the details of model and brand of the arriving car, details of the driver with the vehicle number.

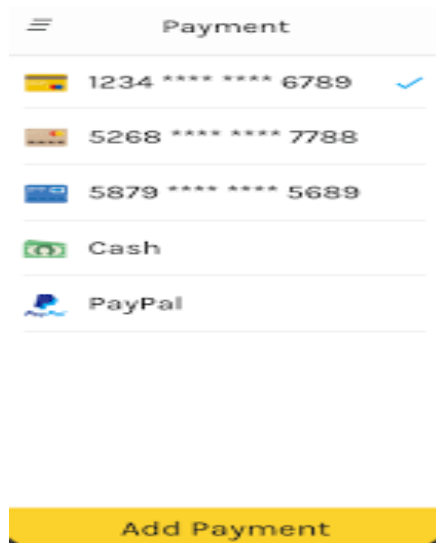


Figure 3.16 Payment

Passengers can pay through any of the various payment options available: Cash, credit/debit cards, online or app wallets.

No.	Ride Id	Driver Name (From)	Rider Name (To)	Rating
1	09350520182011	Rahul Dave	Dharmesh	★★★★☆ (3.5)
2	41391020182111	Rahul Dave	Test User	★★★★☆ (4.5)
3	00490320181110	Mahesh	Mahesh	★★★★★ (5)
4	00490320181110	Mahesh	Mahesh	★★★★★ (5)
5	00490320181110	Mahesh	Rakesh	★★★★★ (5)
6	24280320181110	Mahesh	Rakesh	★★★★★ (5)
7	24280320181110	Mahesh	Avichal	★★★★★ (5)
8	24280320181110	Mahesh	Avichal	★★★★★ (5)

Figure 3.17 Rating

Passengers can rate drivers on a scale of 1 to 5.

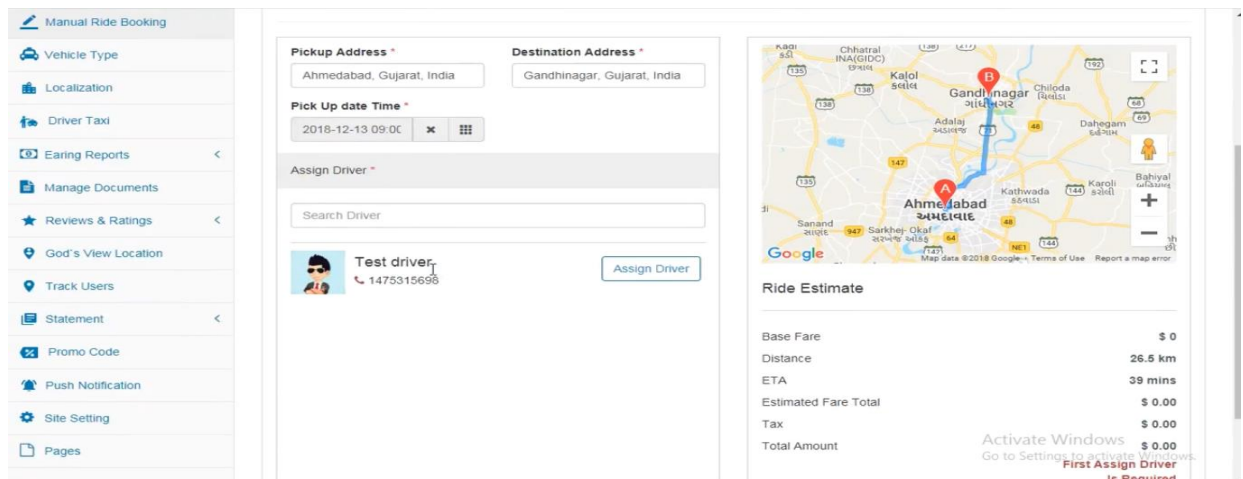


Figure 3.18 Ride Requests

Simple interface to accept or cancel a request and estimated time calculation for reaching the pickup point.

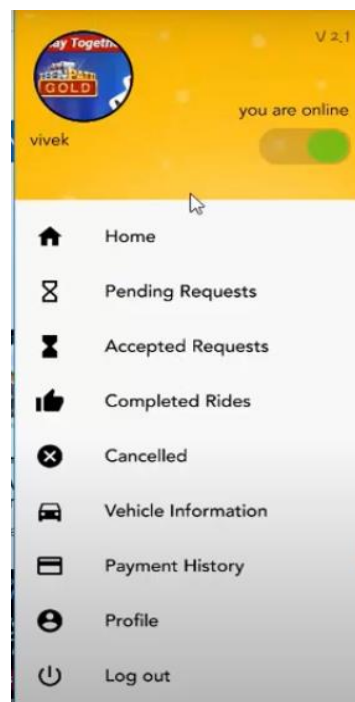
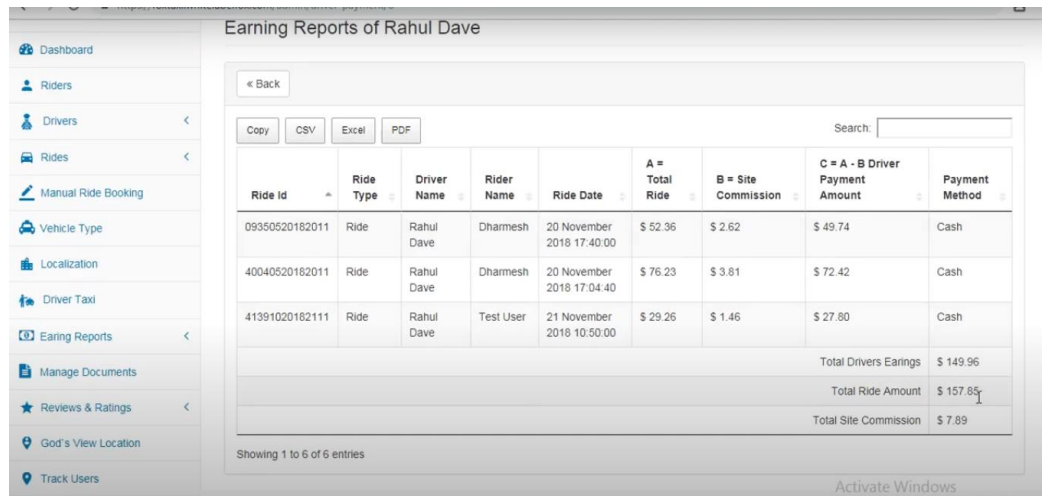


Figure 3.19 Toggle

Drivers can take quick actions like marking arrival, start trip or end trip at the tap of a button.

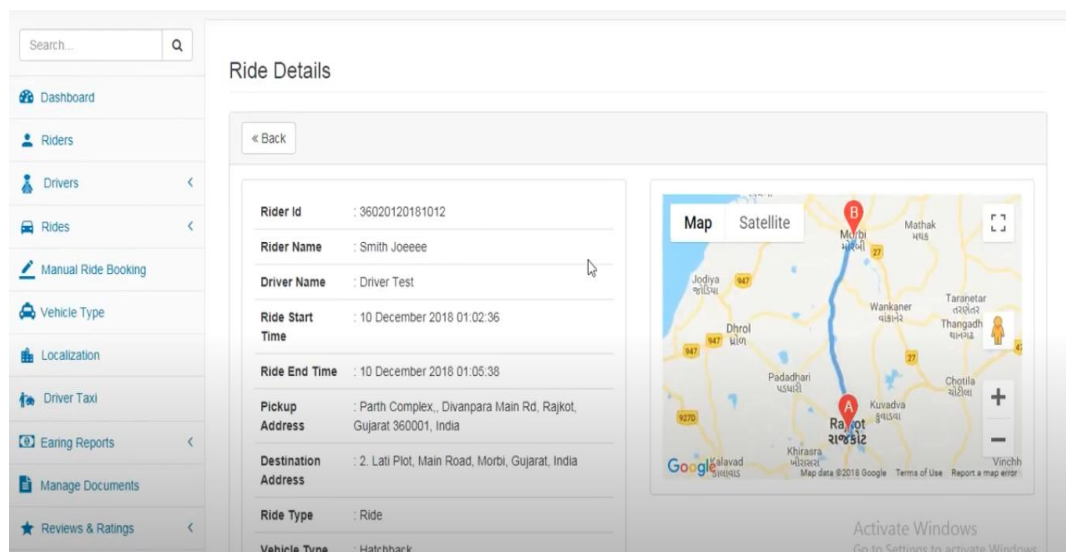


Ride Id	Ride Type	Driver Name	Rider Name	Ride Date	A = Total Ride	B = Site Commission	C = A - B Driver Payment Amount	Payment Method
09350520182011	Ride	Rahul Dave	Dharmesh	20 November 2018 17:40:00	\$ 52.36	\$ 2.62	\$ 49.74	Cash
40040520182011	Ride	Rahul Dave	Dharmesh	20 November 2018 17:04:40	\$ 76.23	\$ 3.81	\$ 72.42	Cash
41391020182111	Ride	Rahul Dave	Test User	21 November 2018 10:50:00	\$ 29.26	\$ 1.46	\$ 27.80	Cash
Total Drivers Earnings								\$ 149.96
Total Ride Amount								\$ 157.85
Total Site Commission								\$ 7.89

Showing 1 to 6 of 6 entries

Figure 3.20 Earnings Dashboard

Drivers can check their average ratings and feedback provided by the passengers.



Rider Id	: 36020120181012
Rider Name	: Smith Joeeee
Driver Name	: Driver Test
Ride Start Time	: 10 December 2018 01:02:36
Ride End Time	: 10 December 2018 01:05:38
Pickup Address	: Parth Complex., Divanpara Main Rd, Rajkot, Gujarat 360001, India
Destination Address	: 2. Lati Plot, Main Road, Morbi, Gujarat, India
Ride Type	: Ride
Vehicle Type	: Hatchback

Figure 3.21 Trip Information

Detailed summary of the trip taken including the taxes, time taken, distance, and a trip map can be retrieved.

Chapter 4

REFLECTION NOTES

Reflection and learning from an internship experience. In experimental learning and internship, the real learning comes after the work term when you have an opportunity to think about what you saw and experienced. Reflecting back about the experience is a key to learning and is definitely not a new idea.

During my work experience at Glisten Project Solutions, I was fortunate enough to have experienced first-hand and learned, many different sides of what goes into project, the general process and detail goes into every stage. Another valuable lesson I have learnt during these 8 weeks, were the different types of work an engineer has to perform, which in turn, have provided me with more insight into the different types of role and responsibilities that I could perform, as a Web Developer.

Reflecting back to my first day in Glisten, it was one of the most memorable day of my time at work. After being introduced to everyone, then our mentor taught us some of the basic concepts of HTML, CSS and JSP and Web Development. And the company people were very open and friendly with the students.

During next weeks, the topic on which the web development was to be deployed was decided. And accordingly, groups were formed and started working on that. Through experience and observation, another valuable lesson that was learnt was that in a project, especially when working in primary stages of certain project, things change quickly and dynamically, thus enabling to understand the concept of continuous learning or lifelong learning. It was found that the workplace itself was relatively diverse from what was observed from farther view.

Working at Glisten, helped the workplace become better as a team and therefore created a good atmosphere in the working environment. The work that needs to be completed varied day-to-day depending on what project was about to begin, or what tender was due. For the development of the application, the group shared the work among each member, that gave a chance to be a part of a team. And finally, the team came up with the application and report. Apart from the project work, during the break, the group was allowed to interact

with one another, developing the feel of good environment, which indirectly contributed to the enhancement of communication skills. Overall, the experience at Glisten Project Solutions was positive. It was felt that it was a great opportunity for a student engineer to have the learn a lot more than one would think. The company was also very satisfied with the work done as an intern for a period of 8 weeks.

EXECUTIVE SUMMARY

This is an attempt to make sure that we as fresh graduates have exposure to the work culture of the industries. As a student of VTU, it is a part of our curriculum for everyone to complete Internship at a good Institute or organization. I was lucky enough to get an opportunity to intern at Glisten Project Solutions Private Limited, Bengaluru.

Initially, I have added information regarding the company which includes its introduction, services, vision and its overall function. Even though the training was for a period of 8 weeks, I was exposed to various technologies as well as applications. The second part of the report talks about Java with Springs and Hibernate and its impact on business, and also details of the hands-on projects I got to do during the training. This environment has given me a good look at the various types of engineering technologies present in today's workplaces.

My future goals extending beyond this internship are to gain valuable classroom knowledge and extra-curricular experience in engineering practices in my final year as an undergraduate. I hope to secure a challenging job as an entry level engineer that will push me to grow and provide me with opportunities to tackle larger projects and handle more responsibility as I grow professionally